

Course 5493A

5 Credits: 4

Program Elective

Source-grounded

Low Power VLSI

□ To know about various short channel effects in MOSFET

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5493A is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5493A.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Microelectronics
Course code	5493A
Course title	Low Power VLSI
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ To know about various short channel effects in MOSFET
- □ To know about various sources of power dissipation in MOSFET
- □ To understand various power reduction techniques in circuits.
- □ To implement logic using various non clocked and clocked design styles.

Course outcomes

CO	Official outcome	Study level
CO1	Identify various short channel effects in MOSFET 15 Understanding Understand various sources of power dissipation	See official syllabus
CO2	14 Understanding in MOSFET Understand various power reduction techniques in	See official syllabus
CO3	14 Understanding circuits Apply various non clocked and clocked design	See official syllabus
CO4	15 Applying styles for logic implementation	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2
CO2	CO2 3 2
CO3	CO3 3 3
CO4	CO4 3 3

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Identify various short channel effects in MOSFET	4	As specified
Module 2	Understand various sources of power dissipation in MOSFET	4	As specified
Module 3	Understand various power reduction techniques in circuits	4	As specified
Module 4	implementation	3	As specified

MODULE 1

Identify various short channel effects in MOSFET

This module is presented from the official syllabus scope for Low Power VLSI. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify various short channel effects in MOSFET

- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain the need for low power circuit design

Explain the need for low power circuit design is an official topic in Module 1 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the need for low power circuit design using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the need for low power circuit design should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the need for low power circuit design means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain the need for low power circuit design
definition/meaning .
steps, application . Technical terms English-
.

Explain the MIS structure 3 Understanding Discuss short channel effects in MOSFET and

Explain the MIS structure 3 Understanding Discuss short channel effects in MOSFET and is an official topic in Module 1 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the MIS structure 3 Understanding Discuss short channel effects in MOSFET and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the mis structure 3 understanding discuss short channel effects in mosfet and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the MIS structure 3 Understanding Discuss short channel effects in MOSFET and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain the MIS structure 3 Understanding Discuss short channel effects in MOSFET and definition/meaning . . . , steps, application Technical terms English-

8 Understanding methods to minimize SCE Describe the design issues with deep

8 Understanding methods to minimize SCE Describe the design issues with deep is an official topic in Module 1 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Understanding methods to minimize SCE Describe the design issues with deep using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 understanding methods to minimize sce describe the design issues with deep should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 Understanding methods to minimize SCE Describe the design issues with deep means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-8 Understanding methods to minimize SCE
Describe the design issues with deep definition/meaning
definition/meaning. definition/meaning, steps, application definition/meaning
definition/meaning. Technical terms English- definition/meaning.

2 Understanding submicron transistor and leakage mechanisms Physics of Power dissipation in MOSFET devices Need for low power circuit design, MIS Structure, Short channel effects (SCE) - Channel Length Modulation, Drain Induced Barrier Lowering, punch through, velocity saturation, surface scattering, Impact ionization, Hot electron effects, Minimizing SCE, Deep submicron transistor design issues, Leakage mechanisms in short channel transistors.

2 Understanding submicron transistor and leakage mechanisms Physics of Power dissipation in MOSFET devices Need for low power circuit design, MIS Structure, Short channel effects (SCE) - Channel Length Modulation, Drain Induced Barrier Lowering, punch through, velocity saturation, surface scattering, Impact ionization, Hot electron effects, Minimizing SCE, Deep submicron transistor design issues, Leakage mechanisms in short channel transistors. is an official topic in Module 1 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding submicron transistor and leakage mechanisms Physics of Power dissipation in MOSFET devices Need for low power circuit design, MIS Structure, Short channel effects (SCE) - Channel Length Modulation, Drain Induced Barrier Lowering, punch through, velocity saturation, surface scattering, Impact ionization, Hot electron effects, Minimizing SCE, Deep submicron transistor design issues, Leakage mechanisms in short channel transistors. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding submicron transistor and leakage mechanisms physics of power dissipation in mosfet devices need for low power circuit design, mis

structure, short channel effects (sce) - channel length modulation, drain induced barrier lowering, punch through, velocity saturation, surface scattering, impact ionization, hot electron effects, minimizing sce, deep submicron transistor design issues, leakage mechanisms in short channel transistors. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding submicron transistor and leakage mechanisms Physics of Power dissipation in MOSFET devices Need for low power circuit design, MIS Structure, Short channel effects (SCE) - Channel Length Modulation, Drain Induced Barrier Lowering, punch through, velocity saturation, surface scattering, Impact ionization, Hot electron effects, Minimizing SCE, Deep submicron transistor design issues, Leakage mechanisms in short channel transistors. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding submicron transistor and leakage mechanisms Physics of Power dissipation in MOSFET devices Need for low power circuit design, MIS Structure, Short channel effects (SCE) - Channel Length Modulation, Drain Induced Barrier Lowering, punch through, velocity saturation, surface scattering, Impact ionization, Hot electron effects, Minimizing SCE, Deep submicron transistor design issues, Leakage mechanisms in short channel transistors. definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Identify various short channel effects in MOSFET

M1.01 Explain the need for low power circuit design 2
Understanding

M1.02 Explain the MIS structure 3
Understanding

M1.03 Discuss short channel effects in MOSFET and 8
Understanding

methods to minimize SCE
Describe the design issues with deep 2
M1.04
Understanding
submicron transistor and leakage mechanisms

Contents:

Physics of Power dissipation in MOSFET devices

Need for low power circuit design, MIS Structure, Short channel effects (SCE) - Channel

Length Modulation, Drain Induced Barrier Lowering, punch through, velocity saturation,

surface scattering, Impact ionization, Hot electron effects, Minimizing SCE, Deep submicron transistor design issues, Leakage mechanisms in short channel transistors.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Understand various sources of power dissipation in MOSFET

This module is presented from the official syllabus scope for Low Power VLSI. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand various sources of power dissipation in MOSFET
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Understanding dissipation

3 Understanding dissipation is an official topic in Module 2 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding dissipation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding dissipation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding dissipation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding dissipation ഓർമ്മപ്പെടുത്തൽ
definition/meaning ഓർമ്മപ്പെടുത്തൽ. ഓർമ്മപ്പെടുത്തൽ ഓർമ്മപ്പെടുത്തൽ, steps,
application ഓർമ്മപ്പെടുത്തൽ ഓർമ്മപ്പെടുത്തൽ. Technical terms English-ഓർമ്മപ്പെടുത്തൽ
ഓർമ്മപ്പെടുത്തൽ.

Explain switching power dissipation

Explain switching power dissipation is an official topic in Module 2 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain switching power dissipation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain switching power dissipation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain switching power dissipation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain switching power dissipation

Explain switching power dissipation definition/meaning. Explain switching power dissipation, steps, application. Technical terms English- Malayalam.

Describe short circuit power dissipation

Describe short circuit power dissipation is an official topic in Module 2 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe short circuit power dissipation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe short circuit power dissipation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe short circuit power dissipation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
Describe short circuit power dissipation
definition/meaning .
steps, application . Technical terms English-
.

Discuss glitching power dissipation 3 Understanding Sources of power dissipation in CMOS (No derivation required) Static Power Dissipation - Leakage Power Dissipation Dynamic Power Dissipation: Switching power dissipation, Short Circuit power dissipation: Short Circuit Current of Inverter, Short circuit current dependency with input and output load, Glitching Power dissipation

Discuss glitching power dissipation 3 Understanding Sources of power dissipation in CMOS (No derivation required) Static Power Dissipation - Leakage Power Dissipation Dynamic Power Dissipation: Switching power dissipation, Short Circuit power dissipation: Short Circuit Current of Inverter, Short circuit current dependency with input and output load, Glitching Power dissipation is an official topic in Module 2 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss glitching power dissipation 3 Understanding Sources of power dissipation in CMOS (No derivation required) Static Power Dissipation - Leakage Power Dissipation Dynamic Power Dissipation: Switching power dissipation, Short Circuit power dissipation: Short Circuit Current of Inverter, Short circuit current dependency with input and output load, Glitching Power dissipation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss glitching power dissipation 3 understanding sources of power dissipation in cmos (no derivation required) static power dissipation - leakage power dissipation dynamic power dissipation: switching power dissipation, short circuit power dissipation: short circuit current of inverter, short circuit current dependency with input and output load, glitching power dissipation should be discussed with attention to

safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss glitching power dissipation 3 Understanding Sources of power dissipation in CMOS (No derivation required) Static Power Dissipation - Leakage Power Dissipation Dynamic Power Dissipation: Switching power dissipation, Short Circuit power dissipation: Short Circuit Current of Inverter, Short circuit current dependency with input and output load, Glitching Power dissipation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Discuss glitching power dissipation 3 Understanding Sources of power dissipation in CMOS (No derivation required) Static Power Dissipation - Leakage Power Dissipation Dynamic Power Dissipation: Switching power dissipation, Short Circuit power dissipation: Short Circuit Current of Inverter, Short circuit current dependency with input and output load, Glitching Power dissipation definition/meaning. definition, steps, application. Technical terms English- Malayalam.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Understand various sources of power dissipation in MOSFET

M2.01	Discuss the sources of static power dissipation	3
Understanding		
M2.02	Explain switching power dissipation	4
Understanding		
M2.03	Describe short circuit power dissipation	4
Understanding		
M2.04	Discuss glitching power dissipation	3
Understanding		
	Series Test - I	1

Contents:

Sources of power dissipation in CMOS (No derivation required)

Static Power Dissipation - Leakage Power Dissipation

Dynamic Power Dissipation: Switching power dissipation, Short Circuit power dissipation:

Short Circuit Current of Inverter, Short circuit current dependency with input and output

load, Glitching Power dissipation

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Understand various power reduction techniques in circuits

This module is presented from the official syllabus scope for Low Power VLSI. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand various power reduction techniques in circuits
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Understanding for power reduction Discuss channel engineering for leakage

2 Understanding for power reduction Discuss channel engineering for leakage is an official topic in Module 3 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding for power reduction Discuss channel engineering for leakage using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding for power reduction discuss channel engineering for leakage should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding for power reduction Discuss channel engineering for leakage means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- 2 Understanding for power reduction Discuss channel engineering for leakage ഓരോന്നിന്റെയും definition/meaning ഉൾപ്പെടെ. ഓരോന്നിന്റെയും steps, application ഉൾപ്പെടെ. Technical terms English-ൽ ഉൾപ്പെടെ.

2 Understanding power reduction Explain the circuit techniques for leakage

2 Understanding power reduction Explain the circuit techniques for leakage is an official topic in Module 3 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding power reduction Explain the circuit techniques for leakage using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding power reduction explain the circuit techniques for leakage should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding power reduction Explain the circuit techniques for leakage means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- 2 Understanding power reduction Explain the circuit techniques for leakage ഹാർഡ്വെയർ ഹാർഡ്വെയർ definition/meaning ഹാർഡ്വെയർ ഹാർഡ്വെയർ. ഹാർഡ്വെയർ ഹാർഡ്വെയർ ഹാർഡ്വെയർ, steps, application ഹാർഡ്വെയർ ഹാർഡ്വെയർ ഹാർഡ്വെയർ. Technical terms English- ഹാർഡ്വെയർ ഹാർഡ്വെയർ.

6 Understanding power reduction Describe the techniques for dynamic power

6 Understanding power reduction Describe the techniques for dynamic power is an official topic in Module 3 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding power reduction Describe the techniques for dynamic power using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding power reduction describe the techniques for dynamic power should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding power reduction Describe the techniques for dynamic power means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- 6 Understanding power reduction Describe the techniques for dynamic power reduction. Define/meaning of dynamic power reduction. Explain the steps, application of dynamic power reduction. Technical terms English- Malayalam.

4 Understanding reduction Power Reduction Techniques Supply voltage Scaling Approaches: Multi VDD and Dynamic VDD Leakage power reduction Techniques : Channel Engineering - Retrograde Doping, Halo Doping, Circuit Techniques - Transistor stacking, VTCMOS, MTCMOS, DTCMOS, Power gating Dynamic power reduction Techniques : Clock gating, Gate sizing, Logic restructuring, Pin swapping Apply various non-clocked and clocked design styles for logic

4 Understanding reduction Power Reduction Techniques Supply voltage Scaling Approaches: Multi VDD and Dynamic VDD Leakage power reduction Techniques : Channel Engineering - Retrograde Doping, Halo Doping, Circuit Techniques - Transistor stacking, VTCMOS, MTCMOS, DTCMOS, Power gating Dynamic power reduction Techniques : Clock gating, Gate sizing, Logic restructuring, Pin swapping Apply various non-clocked and clocked design styles for logic is an official topic in Module 3 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding reduction Power Reduction Techniques Supply voltage Scaling Approaches: Multi VDD and Dynamic VDD Leakage power reduction Techniques : Channel Engineering - Retrograde Doping, Halo Doping, Circuit Techniques - Transistor stacking, VTCMOS, MTCMOS, DTCMOS, Power gating Dynamic power reduction Techniques : Clock gating, Gate sizing, Logic restructuring, Pin swapping Apply various non-clocked and clocked design styles for logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding reduction power reduction techniques supply voltage scaling approaches: multi vdd and dynamic vdd leakage power reduction

techniques : channel engineering - retrograde doping, halo doping, circuit techniques - transistor stacking, vtcmos, mtcmos, dtcmos, power gating dynamic power reduction techniques : clock gating, gate sizing, logic restructuring, pin swapping apply various non-clocked and clocked design styles for logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding reduction Power Reduction Techniques Supply voltage Scaling Approaches: Multi VDD and Dynamic VDD Leakage power reduction Techniques : Channel Engineering - Retrograde Doping, Halo Doping, Circuit Techniques - Transistor stacking, VTCMOS, MTCMOS, DTCMOS, Power gating Dynamic power reduction Techniques : Clock gating, Gate sizing, Logic restructuring, Pin swapping Apply various non-clocked and clocked design styles for logic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 4 Understanding reduction Power Reduction Techniques Supply voltage Scaling Approaches: Multi VDD and Dynamic VDD Leakage power reduction Techniques : Channel Engineering - Retrograde Doping, Halo Doping, Circuit Techniques - Transistor stacking, VTCMOS, MTCMOS, DTCMOS, Power gating Dynamic power reduction Techniques : Clock gating, Gate sizing, Logic restructuring, Pin swapping Apply various non-clocked and clocked design styles for logic definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Understand various power reduction techniques in circuits		
Explain the supply voltage scaling approaches		
M3.01		2
Understanding		
	for power reduction	
	Discuss channel engineering for leakage	
M3.02		2
Understanding		
	power reduction	
	Explain the circuit techniques for leakage	
M3.03		6
Understanding		
	power reduction	
	Describe the techniques for dynamic power	
M3.04		4
Understanding		
	reduction	
Contents:		
Power Reduction Techniques		
Supply voltage Scaling Approaches: Multi VDD and Dynamic VDD		
Leakage power reduction Techniques : Channel Engineering - Retrograde Doping, Halo		
Doping, Circuit Techniques - Transistor stacking, VTCMOS, MTCMOS, DTCMOS,		
Power gating		
Dynamic power reduction Techniques : Clock gating, Gate sizing, Logic		
restructuring, Pin		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

implementation

This module is presented from the official syllabus scope for Low Power VLSI. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: implementation
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

7 Applying implementing Boolean expressions Apply NMOS logic and pseudo-NMOS logic

7 Applying implementing Boolean expressions Apply NMOS logic and pseudo-NMOS logic is an official topic in Module 4 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 7 Applying implementing Boolean expressions Apply NMOS logic and pseudo-NMOS logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 7 applying implementing boolean expressions apply nmos logic and pseudo-nmos logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 7 Applying implementing Boolean expressions Apply NMOS logic and pseudo-NMOS logic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-7 Applying implementing Boolean expressions
Apply NMOS logic and pseudo-NMOS logic ഫലപ്രസാദം definition/meaning
പ്രദാനം. പദപ്രദാനം പദപ്രദാനം, steps, application പദപ്രദാനം പദപ്രദാനം
പദപ്രദാനം. Technical terms English-പദപ്രദാനം പദപ്രദാനം.

4 Applying for function implementation Apply differential cascade voltage switch

4 Applying for function implementation Apply differential cascade voltage switch is an official topic in Module 4 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying for function implementation Apply differential cascade voltage switch using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying for function implementation apply differential cascade voltage switch should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying for function implementation Apply differential cascade voltage switch means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പ്രശ്നം 4 Applying for function implementation Apply differential cascade voltage switch പ്രവർത്തനം definition/meaning പ്രവർത്തനം. പ്രവർത്തനം പ്രവർത്തനം, steps, application പ്രവർത്തനം പ്രവർത്തനം പ്രവർത്തനം. Technical terms English-പ്രവർത്തനം പ്രവർത്തനം.

logic for implementing NAND/AND and 4 Applying NOR/OR circuits Circuit design style - Non clocked circuit design style - fully complementary logic, NMOS and pseudo-NMOS logic, differential cascade voltage switch logic (DCVS) - DCVS NAND/AND, DCVS NOR/OR circuits. Text /Reference: T/R Book Title/Author Kaushik Roy, Sharat C Prasad, Low power CMOS VLSI circuit design, Wiley T1 India, 2000 John P. Uyemura, CMOS Logic Circuit Design, Kluwer Academic Publishers, T2 Newyork, 2001 Ajit Pal, Low-Power VLSI Circuits and Systems, Springer, 2015, DOI : T3 10.1007/978-81-322-1937-8 Abdellatif Bellaouar, Mohamed I Elmasry, Low power digital VLSI design, R1 Kluwer Academic, 1995 Kaushik Roy, Hamid Mahmoodi-Meimand and Saibal Mukhopadhyay, "Leakage R2 control for deep-submicron circuits", Proc. SPIE 5117, VLSI Circuits and Systems, (21 April 2003); <https://doi.org/10.1117/12.498181> Online Resources: Sl. No. Website Link https://www.academia.edu/77767705/Leakage_control_for_deep_submicron_circuits 1 <https://www.scribd.com/document/61476406/paper-C7> 2 <https://www.cerc.utexas.edu/~jaa/vlsi/lectures/3-2.pdf> 3 <https://www.cerc.utexas.edu/~jaa/vlsi/lectures/12-1.pdf> 4 https://vlsi-iitg.vlabs.ac.in/LogicGates_theory.html 5

logic for implementing NAND/AND and 4 Applying NOR/OR circuits Circuit design style - Non clocked circuit design style - fully complementary logic, NMOS and pseudo-NMOS logic, differential cascade voltage switch logic (DCVS) - DCVS NAND/AND, DCVS NOR/OR circuits. Text /Reference: T/R Book Title/Author Kaushik Roy, Sharat C Prasad, Low power CMOS VLSI circuit design, Wiley T1 India, 2000 John P. Uyemura, CMOS Logic Circuit Design, Kluwer Academic Publishers, T2 Newyork, 2001 Ajit Pal, Low-Power VLSI Circuits and Systems, Springer, 2015, DOI : T3 10.1007/978-81-322-1937-8 Abdellatif Bellaouar, Mohamed I Elmasry, Low power digital VLSI design, R1 Kluwer Academic, 1995 Kaushik Roy, Hamid Mahmoodi-Meimand and Saibal Mukhopadhyay, "Leakage R2 control for deep-submicron circuits", Proc. SPIE 5117, VLSI Circuits and Systems, (21 April 2003); <https://doi.org/10.1117/12.498181> Online Resources: Sl. No. Website Link https://www.academia.edu/77767705/Leakage_control_for_deep_submicron_circuits 1 <https://www.scribd.com/document/61476406/paper-C7> 2 <https://www.cerc.utexas.edu/~jaa/vlsi/lectures/3-2.pdf> 3 <https://www.cerc.utexas.edu/~jaa/vlsi/lectures/12-1.pdf> 4 https://vlsi-iitg.vlabs.ac.in/LogicGates_theory.html 5 is an official topic in Module 4 of Low Power VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify logic for implementing NAND/AND and 4 Applying NOR/OR circuits Circuit design style - Non clocked circuit design style - fully complementary logic, NMOS and pseudo-NMOS logic, differential cascade voltage switch logic (DCVS) - DCVS NAND/AND, DCVS NOR/OR circuits. Text /Reference: T/R Book Title/Author Kaushik Roy, Sharat C Prasad, Low power CMOS VLSI circuit design, Wiley T1 India, 2000 John P. Uyemura, CMOS Logic Circuit Design, Kluwer Academic Publishers, T2 Newyork, 2001 Ajit Pal, Low-Power VLSI Circuits and Systems, Springer, 2015, DOI : T3 10.1007/978-81-322-1937-8 Abdellatif Bellaouar, Mohamed I Elmasry, Low power digital VLSI design, R1 Kluwer Academic, 1995 Kaushik Roy, Hamid Mahmoodi-Meimand and Saibal Mukhopadhyay, "Leakage R2 control for deep-submicron circuits", Proc. SPIE 5117, VLSI Circuits and Systems, (21 April 2003); <https://doi.org/10.1117/12.498181> Online Resources: Sl. No. Website Link
https://www.academia.edu/77767705/Leakage_control_for_deep_submicron_circuits 1
<https://www.scribd.com/document/61476406/paper-C7> 2
<https://www.cerc.utexas.edu/~jaa/vlsi/lectures/3-2.pdf> 3
<https://www.cerc.utexas.edu/~jaa/vlsi/lectures/12-1.pdf> 4
https://vlsi-iitg.vlabs.ac.in/LogicGates_theory.html 5 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, logic for implementing nand/and and 4 applying nor/or circuits circuit design style - non clocked circuit design style - fully complementary logic, nmos and pseudo-nmos logic, differential cascade voltage switch logic (dcvs) - dcvs nand/and, dcvs nor/or circuits. text /reference: t/r book title/author kaushik roy, sharat c prasad, low power cmos vlsi circuit design, wiley t1 india, 2000 john p. uyemura, cmos logic circuit design, kluwer academic publishers, t2 newyork, 2001 ajit pal, low-power vlsi circuits and systems, springer, 2015, doi : t3 10.1007/978-81-322-1937-8 abdellatif bellaouar, mohamed i elmasry, low power digital vlsi design, r1 kluwer academic, 1995 kaushik roy, hamid mahmoodi-meimand and saibal mukhopadhyay, "leakage r2 control for deep-submicron circuits", proc. spie 5117, vlsi circuits and systems, (21 april 2003); <https://doi.org/10.1117/12.498181> online resources: sl. no. website link
https://www.academia.edu/77767705/leakage_control_for_deep_submicron_circuits 1
<https://www.scribd.com/document/61476406/paper-C7> 2

<https://www.scribd.com/document/61476406/paper-c7> 3

<https://www.cerc.utexas.edu/~jaa/vlsi/lectures/3-2.pdf> 4

<https://www.cerc.utexas.edu/~jaa/vlsi/lectures/12-1.pdf> 5 [https://vlsi-](https://vlsi-iitg.vlabs.ac.in/logicgates_theory.html)

[iitg.vlabs.ac.in/logicgates_theory.html](https://vlsi-iitg.vlabs.ac.in/logicgates_theory.html) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What logic for implementing NAND/AND and 4 Applying NOR/OR circuits Circuit design style - Non clocked circuit design style - fully complementary logic, NMOS and pseudo-NMOS logic, differential cascade voltage switch logic (DCVS) - DCVS NAND/AND, DCVS NOR/OR circuits. Text /Reference: T/R Book Title/Author Kaushik Roy, Sharat C Prasad, Low power CMOS VLSI circuit design, Wiley T1 India, 2000 John P. Uyemura, CMOS Logic Circuit Design, Kluwer Academic Publishers, T2 Newyork, 2001 Ajit Pal, Low-Power VLSI Circuits and Systems, Springer, 2015, DOI : T3 10.1007/978-81-322-1937-8 Abdellatif Bellaouar, Mohamed I Elmasry, Low power digital VLSI design, R1 Kluwer Academic, 1995 Kaushik Roy, Hamid Mahmoodi-Meimand and Saibal Mukhopadhyay, "Leakage R2 control for deep-submicron circuits", Proc. SPIE 5117, VLSI Circuits and Systems, (21 April 2003); https://doi.org/10.1117/12.498181 Online Resources: Sl. No. Website Link https://www.academia.edu/77767705/Leakage_control_for_deep_submicron_circuits 2 https://www.scribd.com/document/61476406/paper-C7 3 https://www.cerc.utexas.edu/~jaa/vlsi/lectures/3-2.pdf 4 https://www.cerc.utexas.edu/~jaa/vlsi/lectures/12-1.pdf 5 https://vlsi-iitg.vlabs.ac.in/LogicGates_theory.html means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ logic for implementing NAND/AND and 4 Applying NOR/OR circuits Circuit design style - Non clocked circuit design style - fully complementary logic, NMOS and pseudo-NMOS logic, differential cascade voltage switch logic (DCVS) - DCVS NAND/AND, DCVS NOR/OR circuits. Text /Reference: T/R Book Title/Author Kaushik Roy, Sharat C Prasad, Low power CMOS VLSI circuit design, Wiley T1 India, 2000 John P. Uyemura, CMOS Logic Circuit Design, Kluwer Academic Publishers, T2 Newyork, 2001 Ajit Pal, Low-Power VLSI Circuits and Systems, Springer, 2015, DOI : T3 10.1007/978-81-322-1937-8 Abdellatif Bellaouar, Mohamed I Elmasry, Low power digital VLSI design, R1 Kluwer Academic, 1995 Kaushik Roy, Hamid Mahmoodi-Meimand and Saibal Mukhopadhyay, "Leakage R2 control for deep-submicron circuits", Proc. SPIE 5117, VLSI Circuits and Systems, (21 April 2003); <https://doi.org/10.1117/12.498181>

Online Resources: Sl. No. Website Link

https://www.academia.edu/77767705/Leakage_control_for_deep_submicron_cir 1

uits 2 https://www.scribd.com/document/61476406/paper-C7 3

https://www.cerc.utexas.edu/~jaa/vlsi/lectures/3-2.pdf 4

https://www.cerc.utexas.edu/~jaa/vlsi/lectures/12-1.pdf 5 https://vlsi-

iitg.vlabs.ac.in/LogicGates_theory.html ലോജിക് ഗേറ്റ് definition/meaning

ലോജിക് ഗേറ്റ്. ലോജിക് ഗേറ്റ്, steps, application ലോജിക് ഗേറ്റ്

ലോജിക്. Technical terms English-ലോജിക്.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

implementation

Apply fully complementary logic for

M4.01 Applying 7

implementing Boolean expressions
Apply NMOS logic and pseudo-NMOS logic

M4.02 Applying 4
for function implementation

M4.03 Applying 4
Apply differential cascade voltage switch
logic for implementing NAND/AND and
NOR/OR circuits

Series Test - I 1

Contents:

Circuit design style - Non clocked circuit design style - fully complementary logic,
NMOS and pseudo-NMOS logic, differential cascade voltage switch logic (DCVS) -
DCVS NAND/AND, DCVS NOR/OR circuits.

Text /Reference:

T/R

Book Title/Author

Kaushik Roy, Sharat C Prasad, Low power CMOS VLSI circuit design,

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Explain the need for low power circuit design. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the need for low power circuit design*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain the MIS structure 3 Understanding Discuss short channel effects in MOSFET and. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the MIS structure 3 Understanding Discuss short channel effects in MOSFET and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 8 Understanding methods to minimize SCE Describe the design issues with deep. (3 marks)

Answer: Begin with the standard definition or identity of *8 Understanding methods to minimize SCE Describe the design issues with deep*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding submicron transistor and leakage mechanisms Physics of Power dissipation in MOSFET devices Need for low power circuit design, MIS Structure, Short channel effects (SCE) - Channel Length Modulation, Drain Induced Barrier Lowering, punch through, velocity saturation, surface scattering, Impact ionization, Hot electron effects,

Minimizing SCE, Deep submicron transistor design issues, Leakage mechanisms in short channel transistors.. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding submicron transistor and leakage mechanisms Physics of Power dissipation in MOSFET devices Need for low power circuit design, MIS Structure, Short channel effects (SCE) - Channel Length Modulation, Drain Induced Barrier Lowering, punch through, velocity saturation, surface scattering, Impact ionization, Hot electron effects, Minimizing SCE, Deep submicron transistor design issues, Leakage mechanisms in short channel transistors..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain 3 Understanding dissipation. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding dissipation.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain switching power dissipation. (3 marks)

Answer: Begin with the standard definition or identity of *Explain switching power dissipation.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Describe short circuit power dissipation. (3 marks)

Answer: Begin with the standard definition or identity of *Describe short circuit power dissipation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Discuss glitching power dissipation 3 Understanding Sources of power dissipation in CMOS (No derivation required) Static Power Dissipation - Leakage Power Dissipation Dynamic Power Dissipation: Switching power dissipation, Short Circuit power dissipation: Short Circuit Current of Inverter, Short circuit current dependency with input and output load, Glitching Power dissipation. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss glitching power dissipation 3 Understanding Sources of power dissipation in CMOS (No derivation required) Static Power Dissipation - Leakage Power Dissipation Dynamic Power Dissipation: Switching power dissipation, Short Circuit power dissipation: Short Circuit Current of Inverter, Short circuit current dependency with input and output load, Glitching Power dissipation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain 2 Understanding for power reduction Discuss channel engineering for leakage. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding for power reduction Discuss channel engineering for leakage*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding power reduction Explain the circuit techniques for leakage. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Understanding power reduction Explain the circuit techniques for leakage*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 6 Understanding power reduction Describe the techniques for dynamic power. (3 marks)

Answer: Begin with the standard definition or identity of 6 *Understanding power reduction Describe the techniques for dynamic power*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding reduction Power Reduction Techniques Supply voltage Scaling Approaches: Multi VDD and Dynamic VDD Leakage power reduction Techniques : Channel Engineering - Retrograde Doping, Halo Doping, Circuit Techniques - Transistor stacking, VTCMOS, MTCMOS, DTCMOS, Power gating Dynamic power reduction Techniques : Clock gating, Gate sizing, Logic restructuring, Pin swapping Apply various non-clocked and clocked design styles for logic. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding reduction Power Reduction Techniques Supply voltage Scaling Approaches: Multi VDD and Dynamic VDD Leakage power reduction Techniques : Channel Engineering - Retrograde Doping, Halo Doping, Circuit Techniques - Transistor stacking, VTCMOS, MTCMOS,*

DTCMOS, Power gating Dynamic power reduction Techniques : Clock gating, Gate sizing, Logic restructuring, Pin swapping Apply various non-clocked and clocked design styles for logic. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain 7 Applying implementing Boolean expressions Apply NMOS logic and pseudo-NMOS logic. (3 marks)

Answer: Begin with the standard definition or identity of *7 Applying implementing Boolean expressions Apply NMOS logic and pseudo-NMOS logic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Applying for function implementation Apply differential cascade voltage switch. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying for function implementation Apply differential cascade voltage switch*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain logic for implementing NAND/AND and 4 Applying NOR/OR circuits Circuit design style - Non clocked circuit design style - fully complementary logic, NMOS and pseudo-NMOS logic, differential cascade voltage switch logic (DCVS) - DCVS NAND/AND, DCVS NOR/OR circuits. Text /Reference: T/R Book Title/Author Kaushik Roy, Sharat C Prasad, Low power

CMOS VLSI circuit design, Wiley T1 India, 2000 John P. Uyemura, CMOS Logic Circuit Design, Kluwer Academic Publishers, T2 Newyork, 2001 Ajit Pal, Low-Power VLSI Circuits and Systems, Springer, 2015, DOI : T3 10.1007/978-81-322-1937-8 Abdellatif Bellaouar, Mohamed I Elmasry, Low power digital VLSI design, R1 Kluwer Academic, 1995 Kaushik Roy, Hamid Mahmoodi-Meimand and Saibal Mukhopadhyay, "Leakage R2 control for deep-submicron circuits", Proc. SPIE 5117, VLSI Circuits and Systems, (21 April 2003);
<https://doi.org/10.1117/12.498181> Online Resources: Sl. No. Website Link
https://www.academia.edu/77767705/Leakage_control_for_deep_submicron_circuits 1
<https://www.scribd.com/document/61476406/paper-C7> 2
<https://www.cerc.utexas.edu/~jaa/vlsi/lectures/3-2.pdf> 3
<https://www.cerc.utexas.edu/~jaa/vlsi/lectures/12-1.pdf> 4
https://vlsi-iitg.vlabs.ac.in/LogicGates_theory.html. (3 marks)

Answer: Begin with the standard definition or identity of *logic for implementing NAND/AND and 4 Applying NOR/OR circuits* Circuit design style - Non clocked circuit design style - fully complementary logic, NMOS and pseudo-NMOS logic, differential cascade voltage switch logic (DCVS) - DCVS NAND/AND, DCVS NOR/OR circuits. Text /Reference: T/R Book Title/Author Kaushik Roy, Sharat C Prasad, Low power CMOS VLSI circuit design, Wiley T1 India, 2000 John P. Uyemura, CMOS Logic Circuit Design, Kluwer Academic Publishers, T2 Newyork, 2001 Ajit Pal, Low-Power VLSI Circuits and Systems, Springer, 2015, DOI : T3 10.1007/978-81-322-1937-8 Abdellatif Bellaouar, Mohamed I Elmasry, Low power digital VLSI design, R1 Kluwer Academic, 1995 Kaushik Roy, Hamid Mahmoodi-Meimand and Saibal Mukhopadhyay, "Leakage R2 control for deep-submicron circuits", Proc. SPIE 5117, VLSI Circuits and Systems, (21 April 2003);
<https://doi.org/10.1117/12.498181> Online Resources: Sl. No. Website Link
https://www.academia.edu/77767705/Leakage_control_for_deep_submicron_circuits 1
<https://www.scribd.com/document/61476406/paper-C7> 2
<https://www.cerc.utexas.edu/~jaa/vlsi/lectures/3-2.pdf> 3
<https://www.cerc.utexas.edu/~jaa/vlsi/lectures/12-1.pdf> 4
https://vlsi-iitg.vlabs.ac.in/LogicGates_theory.html. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain the need for low power circuit design**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **3 Understanding dissipation**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding for power reduction Discuss channel engineering for leakage**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **7 Applying implementing Boolean expressions Apply NMOS logic and pseudo-NMOS logic**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No. Website Link
- https://www.academia.edu/77767705/Leakage_control_for_deep_submicron_cir
- <https://www.scribd.com/document/61476406/paper-C7>
- <https://www.cerc.utexas.edu/~jaa/vlsi/lectures/3-2.pdf>
- <https://www.cerc.utexas.edu/~jaa/vlsi/lectures/12-1.pdf>
- https://vlsi-iitg.vlabs.ac.in/LogicGates_theory.html

IN-PAGE SEARCH

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